

PROJECT REPORT

CUSTOMER CHURN ANALYSIS

1. Project Title

Customer Churn & Retention Analysis

2. Objective

The main objective of this project is to analyze customer churn.

The project helps to understand:

- Which customers are leaving.
- Which contract has more churn.
- Which payment method has more churn.
- Which internet service has more churn.
- How tenure and monthly charges are related to churn.

3. Dataset

The dataset contains customer information such as:

- Customer ID
- Gender
- Senior Citizen
- Tenure
- Internet Service
- Contract
- Payment Method
- Monthly Charges

- Total Charges
- Churn

4. Tools Used

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Power BI
- Power Query
- DAX
- Google Colab / Jupyter Notebook

5. Data Cleaning

The following steps were performed:

- Loaded the dataset.
- Checked the data.
- Checked missing values.
- Checked duplicate values.
- Corrected data types.
- Removed/handled missing values.
- Created tenure groups.
- Prepared the cleaned dataset.

6. Data Analysis

The following analysis was performed:

- Total customers
- Churned customers
- Retained customers
- Churn rate
- Churn by contract
- Churn by gender
- Churn by internet service
- Churn by payment method
- Tenure analysis
- Monthly charges analysis
- Total charges analysis

7. Python Visualizations

The following charts were created:

1. Churn Distribution
2. Customer Distribution by Contract
3. Churn by Contract
4. Churn by Gender
5. Churn by Internet Service
6. Churn by Payment Method
7. Tenure Distribution
8. Monthly Charges Distribution
9. Tenure vs Monthly Charges

10. Monthly Charges by Churn
11. Total Charges by Churn
12. Churn Rate by Tenure Group
13. Service Usage
14. Correlation Heatmap

8. Power BI Dashboard

Dashboard Name

Customer Churn & Retention Analytics Dashboard

The dashboard contains:

- Total Customers
- Churned Customers
- Retained Customers
- Churn Rate %
- Average Monthly Charges
- Average Tenure

It also contains charts for:

- Churn Distribution
- Churn by Contract
- Churn by Internet Service
- Churn by Payment Method
- Customer Count by Tenure Group
- Monthly Charges by Churn
- Customer Distribution by Contract

- Customer Distribution by Gender
- Tenure vs Monthly Charges
- Customer Details

Slicers used:

- Gender
- Contract
- Internet Service
- Payment Method
- Senior Citizen
- Churn